

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers.

Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks. These include low-level tasks such as natural language inference and paraphrasing, as well as token-level tasks such as named entity recognition and question answering. There are two existing strategies for applying pre-trained language representations to downstream tasks: feature-based and fine-tuning.

Methodology

There are two steps in our framework: pre-training and fine-tuning. During pre-training, the model is trained on unlabeled data over two self-supervised tasks: Masked Language Model (MLM) and Next Sentence Prediction (NSP). For MLM, we mask 15% of the input tokens at random and predict the masked tokens. For fine-tuning, the BERT model is first initialized with the pre-trained parameters, and all parameters are fine-tuned using labeled data from downstream tasks.

Experiments & Evaluation

BERT is evaluated on 11 natural language processing benchmarks, including the General Language Understanding Evaluation (GLUE) benchmark, Stanford Question Answering Dataset (SQuAD v1.1 and v2.0), and Situations With Adversarial Generations (SWAG). BERT_BASE has 12 layers, 768 hidden units, and 110M parameters. BERT_LARGE has 24 layers, 1024 hidden units, and 340M parameters.

Results & Discussion

BERT achieves state-of-the-art results on all 11 NLP tasks. On GLUE, BERT_LARGE obtains an average score of 82.1%, representing a 7.7% absolute improvement over previous state of the art. On SQuAD v1.1, BERT achieves 93.2% F1 score, outperforming human performance.

Limitations

The pre-training objective of Masked LM creates a discrepancy between pre-training and fine-tuning, as the [MASK] token never appears during inference. Additionally, fine-tuning large models requires substantial GPU compute and exhibits instability across random seeds on small target datasets.

Conclusion & Future Work

Recent empirical improvements due to transfer learning with language models have demonstrated that rich, unsupervised pre-training is an integral part of many language understanding systems. Our major contribution is further generalizing these findings to deep bidirectional architectures.